

```

package rooms;
import java.util.Random;

public class RoomTesterArray {

    private static Random rand = new Random();

    public static Room randomRoom(boolean validRoom) {
        Room r = null;
        if (validRoom) {
            int rnum = rand.nextInt(1000 - 100) + 100;
            //try a random number until we get a good one.
            //isValidRoomNo is a static method. We can call from outside
            // as long as it is not private
            while(Room.isValidRoomNo(rnum) == false){
                rnum = rand.nextInt(1000 - 100) + 100;
            }
            r = new Room(rnum, "single", false);
        }
        return r;
    }

    public static void printAll(Room[] arr) {
        for (Room myRoom:arr) { //we will only read the array
            //similar to ArrayLists arrays can be traversed with enhanced for loop
            if(myRoom!=null){
                System.out.println(myRoom);
            }else{
                System.out.println("null");
            }
        }
    }

    public static void printHotel(Room[] arr) {
        for (int i = 0; i < arr.length; i++) {
            if(arr[i]!=null){
                arr[i].showRoom();
            }
        }
        System.out.println();
    }

    public static void main(String[] args) {

        Room[] hotel = new Room[5];

        // we are not using 4 as hardcoded here. We could have done in that way
also        // but here we are trying to impose the reader that we are filling
every spot until the last
        // position. In this way the code is more descriptive. If the array
size change, handwritten
        // number 4 will look weird. The reader need to guess why it is 4.
        for (int i = 0; i < hotel.length-1; i++) {
            hotel[i]= randomRoom(true);
        }

        printAll(hotel);
    }
}

```

```

        boolean randbool = false;
        //we are flipping a coin here
        if (Math.random() > 0.5) { // Math.random creates a floating point
number [0,1)
            randbool = true;
        }

        hotel[4] = randomRoom(randbool);

        System.out.println(Room.getCount() + " rooms are created");

        boolean done = false;
        //reverse traverse
        for (int i = hotel.length - 1; i >= 0 && !done; i--) {
            if (hotel[i] != null) {
                hotel[i].setRoomType("single");
                done = true;
            }
        }

        printHotel(hotel);

    } // end of main
} // end of class

```